

8

IMPROVING NON-NATIVE PRONUNCIATION

Teaching prosody to learners of Spanish as a second/foreign language

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8.1 Introduction

The teaching of pronunciation, which went through periods of indifference, has been gaining traction in recent years (Thomson and Derwing 2014; Derwing and Munro 2015), although many issues, mainly prosodic in nature (see **prosody** and **suprasegmental features**), are still quite neglected, including in the context of Spanish as an **L2** or foreign language (FL) (Hidalgo and Cabedo 2012; Henriksen 2013). On the other hand, children have a precocious sensitivity to prosody and use it to bootstrap grammatical aspects of their language during **L1** acquisition (Höhle 2009). In particular, language-specific perception of intonation seems to be present beginning at four–five months old (Frota and Butler 2018). Moreover, rhythm and intonation are easily noticed when listening to a language, and prosodic features have been considered fundamental factors in the perception of foreign accent, above other grammatical or pronunciation features (Munro 1995; Jilka 2000; Pettorino, De Meo and Vitale 2014).

Through proper control of prosody, a learner not only has a better capacity to produce understandable, natural, and expressive sequences of speech, but also an improved ability to understand speech and to read and speak more fluently across different communicative situations. Unfortunately, being intelligible at the segmental level does not encompass being pragmatically understood, which is crucial since discourse needs to be appropriate to the communicative situation at hand. Indeed, maintaining the prosodic characteristics of the **L1** when speaking another language often leads to misunderstandings, as the same prosodic scheme may be associated with different linguistic or paralinguistic contents in both languages in question (Cruz-Ferreira 1987; Pickering 2001; de-la-Mota 2004; Chen 2005; Mennen 2007).

The development of the strategic competence to adapt to different contexts and interlocutors is part of teaching pronunciation (Cantero 2014). It is not surprising

that this issue has awakened interest in the field of teaching/learning languages, considering that it is also significant from a social approach, that is, it is “critical for L2 pronunciation because it plays a major role in cementing social bonds as a key marker of social identity” (Levis 2016, 154). In sum, prosodic accuracy for learners implies not only more confident and effective interactions with native speakers, but also greater social inclusion.

This chapter addresses the teaching of prosodic features, with relevant aspects of Spanish as a non-native language being motivated through a look at typological variety of languages and research on other aspects of language like variation, discourse, conversation analysis, and pragmatics. The teaching techniques covered include technological means, such as visual representation and manipulation of the physical correlates of prosody, **automatic speech recognition** systems, and multimodal integration. Several specific suggestions for teaching prosody in the classroom are presented as well.

The remainder of this chapter is organized as follows: the prosody of Spanish is described in Section 8.2, the state of research on the acquisition of L2 Spanish prosody is presented in Section 8.3, methods and materials devoted to its teaching are reviewed in Section 8.4, practical guidelines and suggestions are presented in Section 8.5, and finally, some concluding remarks are given in Section 8.6. The chapter’s Appendix contains a sample activity.

8.2 Facing the issue: the prosody of Spanish

Prosody involves units (see **prosodic constituent**) bigger than the **segment** (i.e., suprasegmental), such as the syllable (see Chapter 7), as well as higher units that are, in turn, integrated into even higher units, all of which are part of a prosodic hierarchy. In everyday speech, listeners identify boundaries between speech units, which are marked by relevant cues (i.e., mainly differences in pitch, but also final lengthening, intensity variation, changes in voice quality, and pauses). They segment, structure, and construe speech through phrasing (see **intonational phrase** and **phonological phrase**), thus endowing it with grammatical content. Chunking a stream of speech into units allows us to distinguish sentences such as *Mira la oveja blanca* and *Mira la oveja, Blanca*, which are also disambiguated by punctuation in written texts (Navarro Tomás 1944; de-la-Mota 2009). Although the intermediate level of phrasing has been subject to debate, the intonational phrase is commonly accepted; it has a meaningful intonational contour and is bound by a terminal tonal movement and other phonetic cues (see **phonetics**), such as pauses and lengthening (e.g., ¿*Vendrás al cine?*). Theoretical discussions and many examples of the prosodic hierarchy can be found in Nespor and Vogel (1986), Selkirk (1986), Nibert (2000), Hualde (2003), Aguilar, de-la-Mota, and Prieto (2009, 2009–2014), Prieto and Roseano (2010), and Benet (2012). Phrasing, lexical stress, rhythm, **pitch accent**, and intonation are usually considered the main linguistic attributes of Spanish prosody, although the effects of tempo and voice quality have also received some attention. A review of key prosodic concepts from a theoretical

perspective can be found in Prieto (2003), Himmelmann and Ladd (2008), and Ladd (2008), among others.

The concept of *stress* relates to relative prominence among syllables and is marked through pitch and other **acoustic** cues (see **acoustic phonetics**). Spanish has lexical stress and fairly free lexical accent placement. Although the stressed (i.e., tonic, or most prominent) syllable can be any of the last three of a word (see **oxytone**, **paroxytone**, and **proparoxytone**), or earlier, in verbs with enclitic pronouns (e.g., *can.tán.do.me.las*, see **preantepenultimate stress**), the unmarked (see **marked form**) stress pattern is paroxytone (Quilis 1993, 403). Furthermore, stress can show phonological oppositions in Spanish, as illustrated in the trio **número-numero-numeró**, which shows three different patterns and three different meanings. Unstressed words, which are typically function words (e.g., prepositions, many pronouns), form a unit with a word bearing lexical stress (see **prosodic word**), as in *para la cena* (see Hualde 2011). L1 French and Korean speakers, whose languages base prominence on the phrase level (i.e., accentual phrase) and lack lexical stress, would most likely have difficulty with Spanish's lexical stress system (Jun 2005, Lahoz-Bengoechea 2012).

As for perceived rhythm, languages follow different typological patterns (Pike 1945; Pamies 1999). If a theory of **isochrony** is assumed, Spanish is considered a **syllable-timed** language, which is traditionally described as having stressed and unstressed syllables that are produced with roughly similar duration, and thus, the duration of an utterance is dependent on its number of syllables. Conversely, languages like English, which is considered **stress-timed**, exhibit regular intervals between stressed syllables, meaning, unlike Spanish, the duration of an utterance depends on its number of stressed syllables. Even though rigid isochronous behavior cannot be demonstrated, there are differences in perceived rhythm between types of languages (as stress- and syllable-based languages). In fact, it has been found that linguistic rhythm is related to other phonological attributes, such as syllable structure (see **onset**, **rhyme**, **nucleus [in syllable structure]**, and **coda**), **vowel reduction**, the marking of prosodic heads and edges, and the phonetic ratio of consonantal and vocalic intervals in the speech signal (Gut 2012; Li and Post 2014). Some particular effects of the L1 English rhythmic pattern on L2 Spanish are discussed in White and Mattys (2007). Interestingly, in addition to the differences that would be expected between languages that are typologically so distant, when comparing typologically close languages, some differences also arise; for instance, Italian and Spanish, which are both considered syllable-timed languages, behave similarly but not identically. Gabriel and Kireva (2014) showed that Italian and Porteño speakers (strongly influenced by Italian) transfer some rhythmic properties from Italian to L2 Spanish, such as higher values for both the percentage of vocalic material in the speech signal and the variability of vocalic intervals. As such, it appears that languages are best described as being placed on a continuum between the prototypical timing patterns.

Spanish is considered an **intonational language**, meaning specific changes to melodic contours carry grammatical and pragmatic information. As Quilis points

out, intonation plays several roles: “la entonación es la función lingüísticamente significativa, socialmente representativa e individualmente expresiva de la frecuencia fundamental” (1993, 410). In the **Autosegmental-Metrical** (AM) approach and the Tones and Break Indices (ToBI) transcription framework, melodic movements are considered the result of phonetic interpolation between phonological tonal targets. These targets can be monotonal (i.e., low L or high H) or compounded (e.g., H + L) pitch accents associated with prominent stressed syllables, and * is used to represent the association with a stressed syllable. Boundary tones demarcating intonational phrase edges are represented with %. The Break Index Tier shows indices of cohesion, thus representing levels of prosodic phrasing and degrees of perceived disjuncture, whereas the Tone Tier provides the intonational analysis of an utterance. Usually, main phrasal stress, which is crucial for intonation, falls on a phrase’s last (i.e., nuclear, see **nucleus [in prosody]**) content word and determines the start of a final (i.e., nuclear) contour, which ends at a following boundary tone. This series of movements describes a *nuclear configuration*, which is a reinterpretation of the traditional *tonema* posited by Navarro Tomás (1944).

As for geolectal variation, part of the research conducted within the framework of the project *Atlas Multimedia de Prosodia del Espacio Románico* (AMPER, Martínez Celdrán and Fernández Planas 2003–2016) is described in Congosto Martín (2011). Recent findings show that interrogative patterns support the traditional division into two large areas based on diachrony: *español meridional* or *español atlántico*, as opposed to *español septentrional* (Fernández Planas et al. 2015). Within the AM model, several cross-geolectal studies linked to the project titled *Atlas interactivo de la entonación del español* (AIEEE, Prieto and Roseano 2009–2013) are detailed in Prieto and Roseano (2010) and Hualde and Prieto (2015).

A low phrase boundary is the most common pattern in Spanish statements. This occurs most frequently in a final low nuclear configuration (L* L%) or after a rising–falling pitch accent, where a peak aligns with a stressed syllable and then falls to an L% edge tone (L + H* L%), as illustrated in Figures 8.1 and 8.2.

In prenuclear (i.e., non-final) pitch accents, peaks are often aligned with a post-tonic syllable, meaning they seem to be delayed, or displaced, as also shown in Figures 8.1 and 8.2. Nevertheless, some phonetic variation can be found; for instance, Andean Spanish and Buenos Aires Spanish present unmarked prenuclear peaks aligning within the boundaries of lexically stressed syllables.

To express emphasis or **narrow contrastive focus** at the end of an intonational phrase, a common nuclear configuration is L + H* L% (see Figure 8.3), which contains a rising pitch accent with the peak aligned at the end of the stressed syllable, followed by falling movement to a low edge tone (de-la-Mota 1995; Face 2002).

Another possibility is L* HL% (i.e., rising pitch that achieves its high level later, followed by a final fall), though it is geographically more restricted; however, when both nuclear configurations coexist in a variety, according to Hualde and Prieto: “L* HL% carries a greater emphatic, contradictory force” (2015, 369).

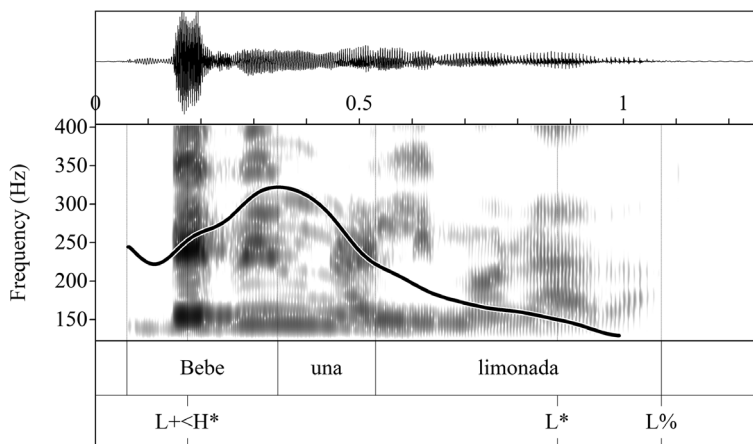


FIGURE 8.1 Acoustic representation and intonational analysis of the neutral statement *Bebe una limonada*, produced with a final low nuclear configuration (L* L%) by a Castilian Spanish speaker.

Source: adapted from Estebas-Vilaplana and Prieto (2010).

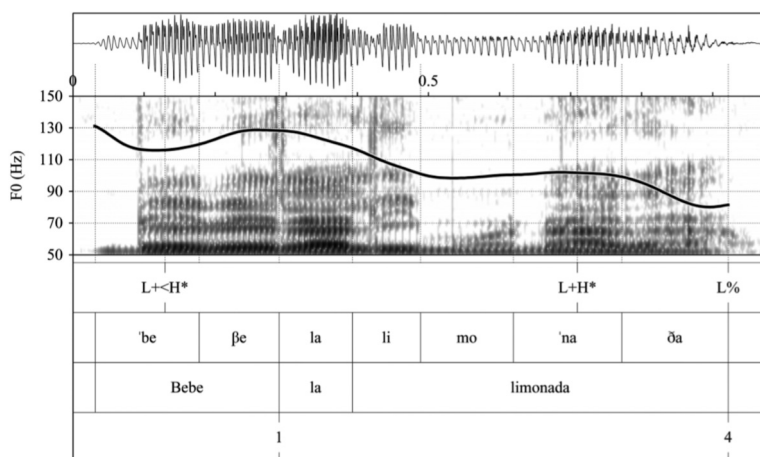


FIGURE 8.2 Acoustic representation and intonational analysis of the neutral statement *Bebe la limonada*, produced with a rising–falling nuclear accent and low boundary tone (L+H* L%) by a Peninsular Spanish speaker.

Source: Hualde and Prieto (2015, 365). This material was originally published in *Intonation in Romance*, eds. Sónia Frota and Pilar Prieto. Reproduced by permission of Oxford University Press, <https://global.oup.com/academic/product/intonation-in-romance-9780199685332?cc=ca&lang=en&#>.

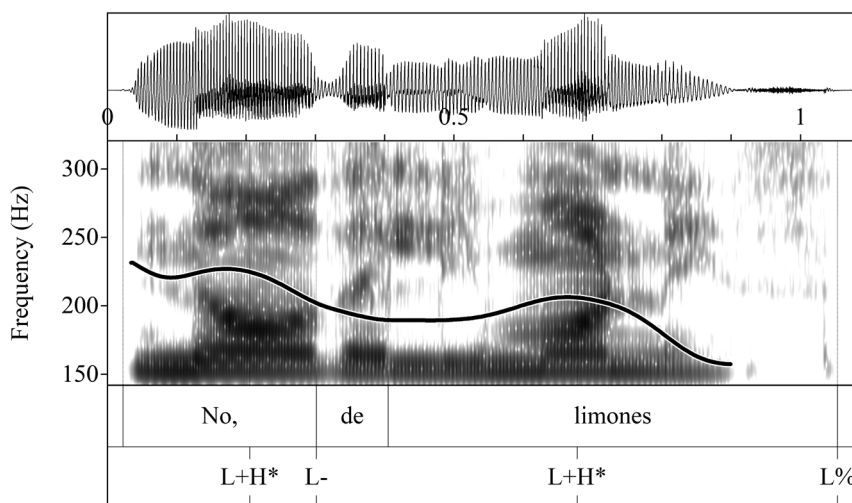


FIGURE 8.3 Acoustic representation and intonational analysis of the narrow focus statement *No, de limones*, produced with an L+H* L% nuclear configuration by a Mexican speaker.

Source: adapted from de-la-Mota, Martín Butragueño, and Prieto (2010).

Moreover, Spanish is quite flexible in terms of word order, which can also be used to convey contrastive focus, or emphasis; for instance, although *MARINA lavó la ropa*, which uses prosody to highlight the subject of the sentence, is a possible way to communicate that it was Marina and not another person who washed the clothes, there are other strategies that could be used (depending on the specific context), such as *Fue Marina quien lavó la ropa* and *Lavó la ropa Marina*. It is interesting to note that, since in English, the usual way to highlight contrastive new information is the nuclear pitch accent, native speakers of English tend to transfer this strategy systematically to Spanish, skipping other possible strategies (Kelm 1987).

Unmarked information-seeking yes–no questions usually have a falling–rising contour in some areas (e.g., Madrid, Mexico) and a circumflex, rising–falling configuration in others (e.g., Caribbean, Buenos Aires) (Prieto and Roseano 2010; Hualde and Prieto 2015). These differences are illustrated in Figures 8.4 and 8.5.

These trends differ from those of other languages; for example, Russian speakers usually end yes–no questions with a drop in pitch (after a sharp rise on the stressed syllable of the nuclear word, Odé 2008; Post 2008), and thus, a falling–rising contour is unexpected. For speakers of tonal languages like Chinese (which, unlike Spanish, use tones for lexical purposes), Spanish interrogative contours can be misleading, since lexical tones affect intonation perception (see Planas 2010; Yuan 2011).

Certain linguistic content (e.g., sentence type or pragmatic nuances) can be conveyed in different geolects by means of different prosodic strategies, that is,

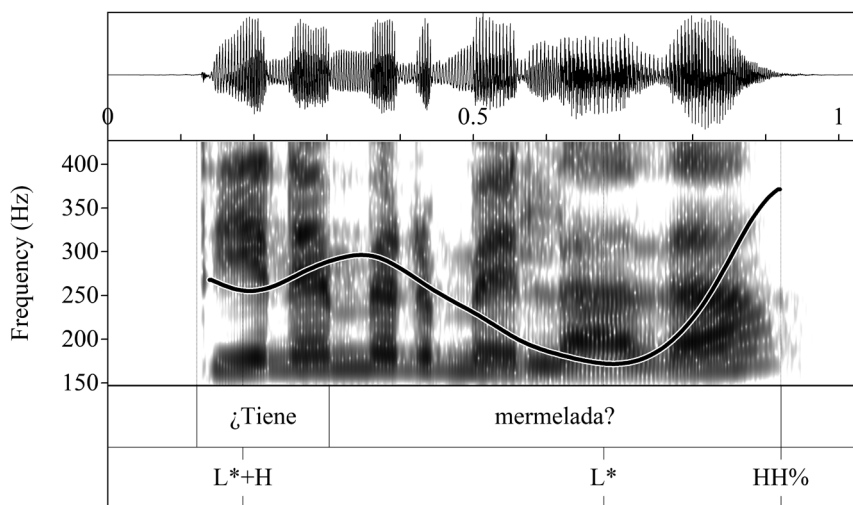


FIGURE 8.4 Acoustic representation and intonational analysis of the information-seeking yes–no question *¿Tiene mermelada?*, produced with a falling–rising nuclear configuration by a Castilian Spanish speaker.

Source: adapted from Estebas-Vilaplana and Prieto (2010).

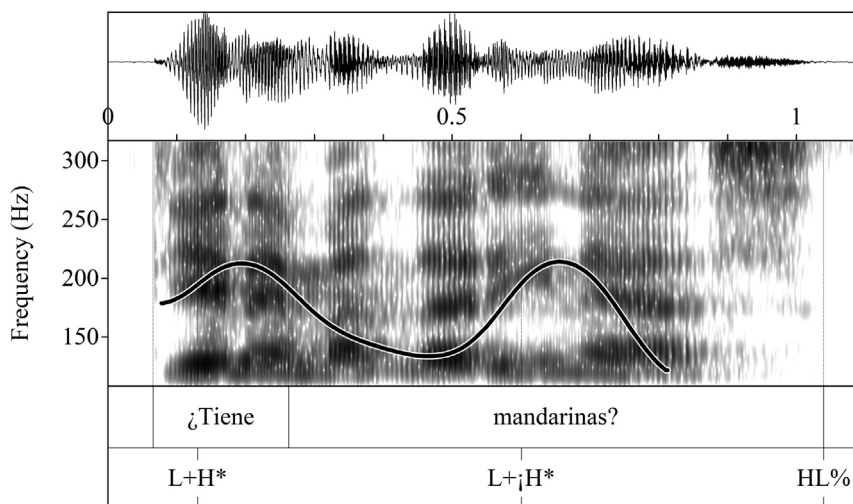


FIGURE 8.5 Acoustic representation and intonational analysis of the information-seeking yes–no question *¿Tiene mandarinas?*, produced with a rising–falling (circumflex) nuclear configuration by a Buenos Aires Spanish speaker.

Source: adapted from Gabriel et al. (2010).

while there are prosodic patterns that seem to be quite widespread, others are restricted to specific regions as instances of microvariation. This is the case with the declarative circumflex (i.e., rise–fall) contour of Mexican Spanish, Argentinian Spanish’s long fall, and Caribbean Spanish’s nuclear hat pattern in yes–no questions (Hualde and Prieto 2015). Conversely, the same pitch accent of a certain intonation contour can be produced differently by speakers from different geolectal varieties in terms of pitch movement and syllabic duration; for example, the main difference between the production of a neutral declarative sentence in Madrid Peninsular Spanish and in Buenos Aires Argentinian Spanish is that in the latter, the nuclear syllable is longer (Estebas-Vilaplana 2010, 167).

As for intralinguistic pragmatic variation, in some cases, a difference between intonational contours is used to convey subtle differences in the meaning of the same sequence of words. As shown in de-la-Mota, Martín Butragueño, and Prieto (2010) for Mexican Spanish, vocatives are commonly described as being pronounced with a stylized vocative chant, or a spoken chant (widely used in other languages like Catalan, Dutch, English, French, and Portuguese), which consists of a high tone associated with the stressed syllable followed by a sustained mid tone (M) in post-tonic syllables, as shown in Figure 8.6; however, there are several possible contours for vocatives, each with its own meaning or meanings, since the contour-nuance relationship is not necessarily univocal. Therefore, as we can deduce from this vocative example, teaching a single intonational contour for a

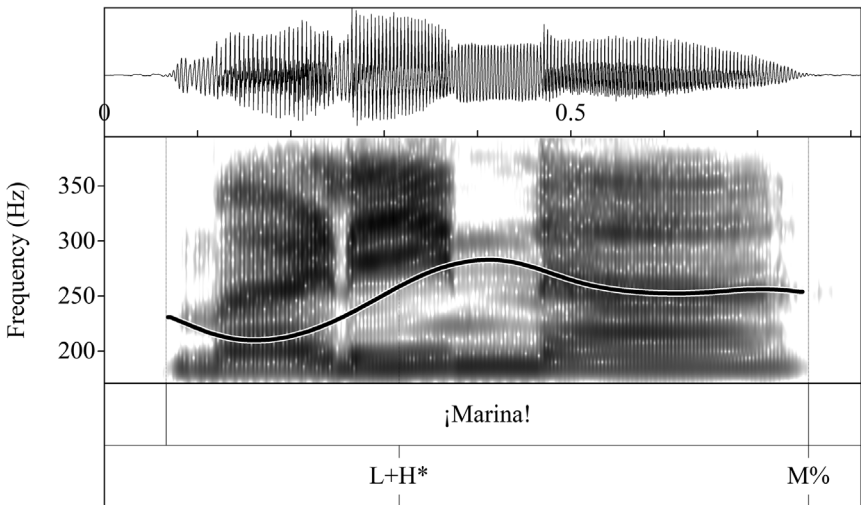


FIGURE 8.6 Acoustic representation and intonational analysis of the tentative call *¡Marina!*, produced with an L+H* M% nuclear configuration by a Mexican speaker.

Source: adapted from de-la-Mota, Martín Butragueño, and Prieto (2010).

given expression does not guarantee that students will be able to express all nuances in the future, that is, it does not provide the learner with an adequate verbal repertoire. Learners' interlanguages may contain a range of contours to express different nuances that also exist in their L1, but this does not assure a one-to-one correspondence with the contour-nuance relationships trying to be acquired in the target language (i.e., successful positive transfer), except in cases of intonational universals (Gussenhoven 2002; Chen 2005). Cases with a lack of one-to-one correspondence could result in learners being misunderstood. An L1 lacking a specific L2 contour (since it expresses the nuance in question in a different way) often triggers negative transfer, where an L1 influenced contour is used inappropriately in the L2. Learners will also have difficulties disambiguating multiple possible meanings when hearing contours used by native speakers. Overall, the communicative and pragmatic challenges mentioned here cause failures that are difficult to avoid without special attention and practice.

Gil Fernández (2007) summarizes some of the overall characteristics of intonation units and pitch variation from a cross-linguistic perspective. She argues that the Spanish intonational phrase is usually of a similar length to that of German, Russian, and English, whereas that of French is shorter and that of Italian is longer. As for pitch range, Gil Fernández claims that there are cross-linguistic differences both in pitch span and pitch level; for instance, French, German, and Spanish have a smaller pitch span than English, whereas Spanish is globally spoken at a low pitch level. For pitch level, she proposes the following scale (from low to high): Spanish < German < English < French < Italian. Perceived pitch scaling also seems to be relevant to evaluate politeness. After performing a perception experiment with short utterances, Estebas-Vilaplana (2014) showed that a high pitch span is judged as natural and polite in English, and as unexpected and overexcited in Spanish, while a low pitch span is considered polite in Spanish, but is commonly viewed as unexpected and rude in English. Yet, when trying to describe the regular behavior of languages from the results obtained in previous work, we must take into account that certain factors hinder interlinguistic comparisons; that is, on the one hand, different measurement standards have been proposed (e.g., for pitch range), and on the other hand, geolectal differences and those related to gender, age, health, and other biological, sociolinguistic and cultural issues have to be considered (see Patterson 2000; Mennen, Schaeffler, and Docherty 2012).

Among other objectives, research on intonation involves finding out the relevant prosodic units for a specific language, the phonological configurations that give rise to the melodic contours of utterances, the distribution of pitch accents, and the grammatical and pragmatic purposes for which configurations are used (de-la-Mota 2009). The following are works that include a detailed description of Spanish prosody as well as its variation: Navarro Tomás (1944), Quilis (1993), Sosa (1999), Prieto and Roseano (2010), Congosto Martín (2011), and the third volume of the *NGRALE* (2011). Other useful descriptions of these issues can also be found in Alcoba and Murillo (1998), Hualde (2005), O'Rourke (2012), Hualde and Prieto (2015), and Prieto and Roseano (2018).

It would be worthwhile for teachers to consider complex typological relations between languages at different prosodic levels, which involves not only the absence or presence of different units, but also each prosodic system in question as a whole. Following Jun (2005), who studies prosodic typology in the AM framework, a first concern is word prosody at the lexical level. It is important to consider tonal languages (e.g., Mandarin and Hausa), stress languages (e.g., Spanish and English), and lexical pitch-accent languages (e.g., Basque and Japanese), while also taking into account that languages can be specified for more than one of these features or lack all of them. A second question relates to post-lexical prosody (i.e., intonational pitch accent, phrase accent), which cannot be directly predicted from lexical prosodic features. Similarly, prosodic units higher than the word do not depend on the timing unit of a language (conventionally, syllable-timed like Spanish, stress-timed like English or mora-timed (see **mora**) like Japanese); however, these units define perceived rhythm. Finally, the same prosodic units (i.e., a certain pitch contour) can be used with different distributions or with different linguistic purposes in different languages. From the perspective of teachers, considering cross-linguistic typology could be daunting, especially in cases where they have a large and linguistically diverse group of students, because it entails taking into account the L1s (and other languages previously acquired) of students when selecting the most appropriate types of exercises.

8.3 Remarks on the acquisition of prosody in L2 Spanish

For several decades, it has been assumed that the L1 system has a crucial influence on the acquisition of an L2. Approaches to phonological acquisition with different conceptions of the native perceptual framework have given rise to several L2 speech learning models that take into account cross-language speech perception and production (or processing), such as the Perceptual Assimilation Model (PAM, Best 1995) and PAM-L2 (Best and Tyler 2007), the **Speech Learning Model** (SLM, Flege 1995), the Native Language Magnet Model Expanded (NLM-e, Kuhl et al. 2008), which is based on previous versions of the model, and the Ontogeny and Phylogeny Model (OPM, Major 2001), which is a revised version of the Ontogeny Model (OM, Major, 1987) (for further discussion of these models, see Chapter 12 of this volume). One of the crucial issues these models identify is similarity; whereas very different sounds from those of the L1 are easier to discriminate and acquire, sounds similar to those of the L1 are more problematic. Some research on segmental information shows that higher-order generalizations settled during L1 acquisition can explain the perceptive behavior of learners better than mappings of non-native sounds onto native-language phonetic categories (Pajak and Levy 2014). The Prosodic-Learning Interference Hypothesis (Tremblay et al. 2016) also addresses the degree to which a prosodic system similar to that of an L2 can be more of an obstacle than an aid. Furthermore, Mennen (2015) presents the L2 Intonational Learning theory (LILt), where she analyzes L1–L2 transfer along four dimensions: 1. Systemic, which involves the phonological inventory and distribution of

categories, 2. Realizational or phonetic, 3. Semantic or functional, and 4. Frequency of use of the categorical elements in the L1 and L2. These dimensions inform which kinds of intonational transfers can and cannot occur in any language, thus generating predictions as to the relative difficulty of production and perception tasks. In fact, although pitch and duration are used across languages and can be considered universal prosodic cues, Kim, Broersma, and Cho (2012) show that their specific use is modulated by individual prosodic phonologies, which also ends up guiding the learning of unfamiliar languages. Some recent works, like those by Colantoni, Steele, and Escudero (2015) and Derwing and Munro (2015), offer an overview of understanding non-native speech from a theoretical and applied perspective.

It has been about 30 years since the acquisition of Spanish as an L2 started to be studied from a variety of methodological and theoretical perspectives, such as functionalist, variationist, generative, psycholinguistic, and cognitive (see Geeslin 2013). The study of prosody has given rise to well-known theoretical models, such as the British and the American Schools, the Dutch or IPO model, the Aix-en-Provence program, the discourse-intonation approach, and the AM framework of intonation, among others (Prieto 2003; Hidalgo 2014). Still, theoretical work on acquisition that specifically takes into account L2 Spanish prosody remains sparse (see Cortés 2002a, 2002b; Radel 2008). After a detailed review of suprasegmental phenomena in L2 Spanish, Henriksen (2013) concludes that plenty of future research avenues are available to researchers since we are still far from comprehending the hierarchy of prosodic acquisition. Without a comprehensive understanding of the acquisition process of each linguistic phenomenon of interest, it is very difficult to create appropriate pedagogical interventions; however, in language teaching, it is inevitable to act without being able to wait for complete, firm, and proven results (Martín Peris 1996). Luckily, there are aspects about which we have enough information to make some substantiated claims.

8.4 Teaching prosody: “To be or not to be, that is the question”

Not long ago, Carbó et al. stated: “En el campo de la enseñanza del español como lengua extranjera faltan materiales que respondan a planteamientos comunicativos para el aprendizaje de la pronunciación y materiales metodológicamente adecuados para emplearlos en la corrección fonética” (2003, 20). Currently, although the teaching of pronunciation has piqued more interest, and several valuable works on pronunciation have been published (see Thomson and Derwing 2014; Derwing and Munro 2015), prosody is largely overlooked in L2 teaching and is still not addressed accurately in many L2 Spanish textbooks. As such, appropriate materials are still needed in general, and need to be adjusted to different learner levels in particular (Santamaría Busto 2007; Ballesteros 2008; Hidalgo and Cabedo 2012; Henriksen 2013). Despite a relatively recent increase in the use of speech technology for the teaching of Spanish pronunciation (see Black 2007; Levis 2007; Eskenazi 2009; Campillos Llanos 2010; Chun 2013; Garrido 2017, Chapter 10 of this volume),

prosody has not been in the spotlight. A detailed review of 23 computer-aided pronunciation teaching applications (see **Computer-Assisted Pronunciation Training**) for Spanish as an L2/FL in Escudero and Carranza (2015) shows that such activities can include: 1. The use of previously recorded voices to improve perception, 2. Listening to synthesized speech, 3. Recording and listening to learners' voices, and 4. Recognition of learners' speech using previously produced statements or open domains, such as dialogue and dictation systems. Yet, the acquisition of prosody (addressed in varying degrees of detail and depth) is considered in only six of these 23 applications: *Connected Speech*, *EyeSpeak*, *Pronunciation Coach*, *Pronunciation Power*, *Tell Me More*, and *Transparent Language*. This implies that there is still plenty of room for improvement in the field. Although the shortage of published materials does not necessarily imply that the issue is not addressed in day-to-day work, it is certainly a sign of an understudied and underemphasized discipline that is difficult for teachers to tackle (see Boquete Martín 2014).

In addition to using computer-aided pronunciation teaching applications, teachers can use free-access programs that allow students to analyze their voices, among other educational materials available on the Internet; however, it may not be clear to designers to what extent these materials might be needed and used in L2 learning, and teachers may not know how to use them properly and effectively. In addition, there are still some teachers who believe that prosody is too difficult to learn (if not impossible) or, conversely, that it will be learned naturally without any explicit instruction, as in L1 acquisition. Even if teachers are motivated, sometimes they have little training in the phonetics and pedagogy of pronunciation and experience a lack of confidence when it comes to prosody instruction (for a discussion of related topics, see Chapter 15 of this volume). Linguists have sometimes unwittingly contributed to this feeling by presenting the analysis and teaching of prosody as matters of great complexity (see Cortés 2002a, 2002b; Trouvain and Gut 2007; Derwing and Munro 2015). Furthermore, the scarcity of materials is not surprising given that research on the description of prosodic features, prosodic typology, and the acquisition of L2 Spanish prosody is in a state of flux, with many outstanding issues remaining unresolved. Additionally, the vast majority of cross-linguistic trends cited in previous research are based on controlled, laboratory speech. Therefore, despite increased effort to change this in recent years, it remains necessary to expand analyses to look through the lens of natural communicative situations (Face 2010).

For years, interest in effectively guiding the learning of L2s has led to the emergence of different teaching methods, some of which hardly touch upon pronunciation (if at all). Within the Grammar and Translation Method, pronunciation was not given much attention, while the Phonoarticulatory and Audiolingual Methods placed it in the foreground. Audiolingual Methods (i.e., direct, situational, and audio-oral) assigned high priority to listening and speaking skills, as they were influenced by structural linguistics and behavioral psychology, but the teaching process was mainly based on the sequence “listen, repeat and memorize,” which is, in fact, impossible to accomplish due to the “phonological deafness” of learners (in Trubetzkoy's 1939 sense). Indeed, the problem lies in the inability to properly

perceive the sounds of the target language because the L1 acts as a filter. This idea is developed within the Verbotonal method, which was first designed for the hearing-impaired, but was extended by Renard (1979) to language teaching. Within this method, which is based on naturalness and the absence of intellectualization, prosody is considered as the basis for the acquisition of segmental units, and a close relationship between speech and gesture is established (see Renard 1979; Intravaia 2000; Billières et al. 2013). Adopting the Contrastive Prosody method, Missaglia (1999) demonstrated the effectiveness of prosody-centered pronunciation training and recommended prioritizing prosody in L2 acquisition.

During the final quarter of the twentieth century, the Cognitive Approach (influenced by Generative Linguistics, see **Generative Phonology**) gave priority to grammar and focused on enhancing the creative capacity of language, while the Communicative Approach placed emphasis on the development of the communicative competence of learners, including task-based language teaching and project work. A certain interpretation of this model led to the misleading thought that pronunciation was not crucial (since it was considered just a matter of form rather than function), but some studies have endorsed the convenience of this approach with respect to prosodic meaning. McKinnon (2016), for example, in a study on how L2 Spanish speakers distinguish imperatives from declaratives using intonation, showed that the instruction of intonational contours can be improved through a task-based approach. For a more complete review of the approaches, designs, and procedures mentioned in this chapter, see Cortés (2000), Gil Fernández (2007, 2012), Paredes (2013), and Santamaría Busto (2013), among others.

Despite the obvious differences between all the methods previously touched upon, the Verbotonal and Communicative approaches are compatible and complementary, and can be combined when designing teaching strategies and materials (Padilla 2007). Cognitive and Communicative approaches and explicit and implicit approaches to grammar are combined in the top-down PACE Model (summarized in Adair-Hauck and Donato 2002) and in its extension, the S-PACE Model (see Chapter 3 of this volume). This model could be applied to prosody, while potentially incorporating Verbotonal and gestural techniques, since it uses authentic speech samples and works on how forms express intended meanings in interpersonal communication.

Some activities proposed within other frameworks mentioned in this section can also be useful if they are brought together and integrated during the process of learning prosody. In order to do this effectively, it is necessary for teachers of L2s to thoroughly evaluate their theoretical options and to stay up-to-date in their training (see Chapter 15). A critical review of methods in L2 teaching, which is also relevant to the field of pronunciation, has led to an emerging and still evolving post-method pedagogy, where teachers are not constricted, but rather fully autonomous since they “theorize from their practice and practice what they theorize” (Kumaravivelu 2006, 184).

Once the topic of theoretical approach is resolved (assuming it is), the question arises as to which pronunciation norm should be taught in the classroom (see Chapters

9 and 14 of this volume). Teachers face this problem right from the beginning levels of language classes. Spanish, which is widely spoken, has a polycentric (or pluricentric) nature, with several influencing centers and, as advised by Carbó et al. (2003, 162), recommendations on pronunciation should specify whether the scope of use is general or restricted to a certain geographical area. Finally, it is recommended that variation be introduced gradually, most likely in higher levels of classes (Moreno Fernández 2010).

8.5 From theory to practice: several tips

Some guidelines and suggestions on how to approach prosody when teaching Spanish as an L2/FL are provided below (for general recommendations regarding teaching pronunciation, see Chapter 15). Designers of pedagogical materials and teachers can find other specific activities and proposals to improve the prosody of speakers of Spanish as an L2/FL in Poch Olivé (1999), Cortés (2002), Iruela (2004), Santamaría Busto (2007), Gil Fernández (2007, 2012), Ballesteros (2008), Hidalgo and Cabedo (2012), and Padilla (2015), among others. The Appendix contains a sample activity that integrates various concepts discussed in this section.

8.5.1 *Don't shy away from phonetics*

Although teachers of Spanish as an L2/FL are not required to be specialists in any area of linguistics, including phonetics and phonology, it is recommended that they: 1. Have enough theoretical knowledge to recognize universal categories and patterns, and to deal with challenges associated with similarities and differences between learners' L1 and the target language, and 2. Know how to integrate a range of technical and practical skills suitable to developing the pronunciation abilities of students (Gil Fernández 2007; Hidalgo and Cabedo 2012; Hidalgo 2015; Poch Olivé 2016).

8.5.2 *Be cognizant of linguistic variants and didactic speech*

For teachers, using the normative pattern of their linguistic variety is the most straightforward classroom approach, but this does not always coincide with the specific Spanish norm that they feel should be encouraged in their sociocultural situation, particularly in the early stages of learning (see Chapter 9). It is also quite common for teachers to be non-native speakers of Spanish, which could potentially result in a distinct source input variety that is influenced by features of their L1. Moreover, teachers often use a specific speech style with their students, known as *classroom speech* or *teacher talk*, in order to lighten cognitive demand, which is very appropriate in an L2 context. This practice affects the prosodic model that it is transmitted from teacher to student, superimposing a special use of emphatic prosodic strategies that includes “pitch range expansion, early peak alignment, increased time and extent of pitch rises, less pitch reduction, and stress addition” (Rao 2011, 493). These features, similar to those found in contrastive focus, are also present in

other communicative situations, such as broadcasting (de-la-Mota and Rodero 2011). As in language teaching and learning, these situations are also constrained by the cognitive needs of the addressee, and serve as an example of a restricted use of speech style, where particular strategies are employed that are not common in spontaneous conversations and other daily interactions.

8.5.3 Provide natural speech models

When the target language is actually spoken during classes (as it should be, regardless of learner level), teachers become the primary linguistic model for students. Without completely excluding interlanguage (which can strengthen social ties within the classroom and promote conscious knowledge, García and Sylvan 2011), teachers must regularly use the language being taught in class. The classroom should be based on authentic linguistic samples and genuine and varied L2 input that shows language in context and reflects all types of variation. As Navarro Tomás states: “es, por supuesto, en la casa, en la oficina, en el comercio o en las tertulias y reuniones de sociedad donde la entonación se produce con espontaneidad y soltura” (1944, 216). These speech samples can be obtained from digitized resources or the Internet. Indeed, it is relatively simple to access naturally occurring speech data from interactions in different geolectal, sociolinguistic, and stylistic varieties, and in different communicative situations. Relevant data can also be obtained from prosodic corpora, which have been designed while taking into account situational variation and other sociolinguistic factors. Some examples are Glissando (Garrido et al. 2013) and intonation atlases like AIEEE and AMPER (mentioned in Section 8.2), all of which can help illustrate how different pragmatic nuances are conveyed in different varieties; for instance, an interrogative can have an imperative rather than a question interpretation (e.g., *¿Os queréis callar?*), but an interrogative can also be a counter-expectational question when uttered with surprise, astonishment or even incredulity (e.g., *¿Necesitas otra manta?*). Working with these kinds of nuances means paying attention to the relation between prosody and discourse, and providing students with natural speech situations that show how prosody can be affected by informational content, illocutionary value and speakers’ intentions, the structuring of large discourse units, turn-taking strategies, and discursive resets, among other factors (Wichmann 2000; Wennerstrom 2001; Escandell-Vidal 2011; Wharton 2012). In the classroom, several kinds of tasks can be designed to become familiar with and to practice these nuances associated with prosody; for example, guessing the intended purpose of a speaker and answering accordingly or persuading an audience with a self-created advertising slogan in a role-playing task.

8.5.4 Assess theoretical approaches first, then choose or combine methods

It can be useful to take advantage of the most convenient tips of different methods (e.g., Phonoarticulatory, Audiolingual, Cognitive, Communicative, Verbotonal,

and PACE) provided that the resulting selection is consistent with the theoretical perspective that each teacher considers appropriate for a certain teaching/learning process. Some tasks presented here will reinforce conscious learning, so it is necessary to decide when and how to introduce them and if they can be blended with other approaches. Despite what is suggested in some methods (which exclude all formal teaching in pronunciation acquisition), when needed, even clear and short explanations can be helpful. While dealing only with the awareness of the learner (as in the Phonoarticulatory Method) or using just listening-and-(trying to)-repeat exercises (as in the Audiolingual Method) are not recommended, it could be useful to combine some of these techniques with others. Ghazi-Saidi and Ansaldo (2017), for example, used neuroimaging techniques and word repetitions to show that repeated practice induces neuroplasticity at the network level (i.e., changes to connectivity in the brain), which is modulated by L1-L2 distance. Further studies of this sort are required, but blending perspectives has obvious implications for L2 teaching. Moreover, though repetition tasks in L2 acquisition are controversial, they can be used in a meaningful way within a communicative context, which can result in improved L2 intonation (Trofimovich and Gatbonton 2006).

8.5.5 Pave the way for learning prosody

A productive way of preparing students to learn prosody is through a pre-phonetics phase that deals with only vocal and ear training exercises. As for perception (which is the basis production), familiarity with steady and dynamic tones can be achieved by means of identification and discrimination tasks, including the ordering of pairs of sounds, and counting and locating sounds. These types of language-independent ear training activities are readily available on the Internet (e.g., Aguilar, de-la-Mota, and Prieto 2009).

Directed phonoarticulatory training (including reading aloud, in the case of advanced learners) can also help with proper overall production (Díaz Pérez 2007). This means developing an awareness of articulations (see **articulatory phonetics**) or language-specific articulatory settings (i.e., the articulatory adjustments that characterize a whole phonetic system in a particular way). This goal can be achieved through exercises that promote sensitization and familiarization with Spanish's articulatory settings, such as a high degree of articulatory tension, **modal phonation**, central-**front resonance**, poor **labialization**, substantial activity of the apex of the **tongue**, sharp jaw displacement, and a low degree of nasality (see **nasal murmur**) (Gil Fernández 2007). General production exercises about breathing, voice quality, modulation, sound quality, resonance, clarity of pronunciation, and diction must be combined with those devoted to improving more specific prosodic issues.

8.5.6 Know learners, contrast systems

It is impossible to propose a set of exercises suitable for everyone since learner profiles demonstrate tremendous variation (see Santos Rovira 2012). The L1 often acts

as a filter that causes L2 input to be reinterpreted in terms of the L1 system (see Chapter 13 of this volume). Therefore, in order to “make a diagnosis” and predict learners’ “error system” (i.e., those most likely to occur), teachers need to be familiar with learners’: 1. L1 and geolectal varieties, and their linguistic similarities and differences when compared to the target language, 2. Competence in the target language, and 3. Degree of exposure to other languages or varieties. To illustrate these points, a useful reference is Blanco Canales and Nogueroles López (2014), who examine pronunciation errors made by L2 Spanish speakers who are native speakers of 11 different languages.

It is also advisable for teachers to know how prosodic universals work (Gussenhoven 2002, Chen 2005) because they can simulate cases of positive transfer from the L1 (since nothing new has to be learned). The markedness of each specific phenomenon is considered to be a factor that affects the difficulty of learning, so if an aspect that distinguishes two languages is more marked in the L2 than in the L1, it will be harder to learn (see Rutherford 1988). Nevertheless, as predicted by the Prosodic-Learning Interference Hypothesis (Tremblay et al. 2016), similar prosodic systems (i.e., microvariation) could also trigger more trouble than having very different prosodies, at least within the realm of learning certain cues related to speech segmentation; for example, using a pitch rise to identify word-final boundaries in French. For teachers, the biggest challenge is working with heterogeneous learners. Despite the difficulties involved, a helpful strategy for teachers is to divide their classes into smaller groups of learners from similar language and regional backgrounds when addressing specific prosodic issues.

8.5.7 Include prosody from day one

In a top-down approach, improving prosody has to be a goal right from the beginning of a term because it improves early learners’ self-confidence and helps prevent the development of inappropriate communicative habits; however, it should also be incorporated in teaching advanced learners so that they can move further toward their L2 communicative goals. At the beginning of a term, it is recommended that teachers implement diagnostic tasks in order to gain a better sense of the needs of students (weaknesses and strengths). Once the teaching process begins, both intelligibility and naturalness should be focused on.

8.5.8 Develop each prosodic aspect in a precise order

The order of teaching interventions depends on the results of diagnostics of learners’ abilities, although the progression recommended here accounts for not only individual differences, but also universal criteria. Some recommendations include: 1. Begin with perception before moving on to production, 2. Start with implicit (i.e., non-verbal) teaching before moving on to explicit (i.e., verbal) teaching, 3. Encourage work in groups before moving to the individual level, 4. Move from oral and audiovisual to writing, 5. Put prosody before sounds in a top-down approach and

move to **bottom-up** techniques later, 6. Start with short intonation units and continue working on prosody up to the level of discourse, and 7. Introduce pragmatic nuances later and work on them steadily (Chun 2002; García Andrevia 2014). Regarding the last point, it is still not clear when variation in pragmatic nuances should be addressed. As for markedness in intonation, according to **Relevance Theory** and L1 acquisition processes, Lahoz-Bengoechea (2007) posits that marked contours should be presented first. This is a crucial issue since contours that are widely used across languages (and perhaps universally) are typically presented first. Such a shift would entail rethinking the order in which intonation patterns are introduced when designing pronunciation syllabi. Lahoz-Bengoechea (2007) considers that, since unmarked contours can be used with a greater number of nuances, learning them first could be an obstacle to learning the specific interpretations of marked contours. Along these lines, he argues that instead of teaching absolute interrogatives with a final rise first (as in the sample activity in the Appendix of this chapter), it would be advisable to start with other contours with more restricted interpretations; however, some works on L1 acquisition (e.g., Thorson et al. 2015) have shown that, even before acquiring the ability to produce two words utterances, Spanish children produce rising yes–no questions with a form–meaning association similar to that of child-directed speech. Therefore, it appears that further evidence is needed and more precise clues as to the best order of introducing each prosodic aspect are still under discussion (Mennen 2015). Moreover, L2 prosodic acquisition seems to be multisystemic in nature; for example, the prosody of interlanguage demonstrates that the acquisition of rhythm emerges gradually rather than uniformly since the specific properties involved are acquired at different acquisitional stages (Lahoz-Bengoechea 2012; Li and Post 2014).

8.5.9 Consider using chunking

Other useful approaches to teaching intonation involve the lexicon and pragmatics, where students do exercises with meaningful, prefabricated “chunks” of language (Seidlhofer and Dalton-Puffer 1995; Travalia 2007). This can also be done by taking into account the specific rhythmic pattern and prosodic units of Spanish, and by exploiting the differences between these properties and those of learners’ L1. Later, these chunks can be integrated into larger linguistic units in contextualized speech and used in (semi)spontaneous activities in order to become fully meaningful since this is how natural discourse works (Chun 2002).

8.5.10 Incorporate literature and songs

Teachers are encouraged to create an entertaining environment in which learners are protagonists. Prosodic phenomena can be illustrated and practiced through literature and songs, which are also helpful for fluency (see Chapters 6 and 7), by paying attention to **resyllabification**, stress, phrasing, boundaries (including pauses), timing, and intonation. Examples can come from traditional and popular

songs, but also from more current genres with social impact, such as rap (from hip-hop culture), which is based on a type of rhythm that combines gestural beats and words with individual flow. Translating a poem or a rap song is a task grounded in phonology. Such activities enhance cultural and social integration and help students discover the rhythmic constraints of typologically different languages. Furthermore, while the teaching of pronunciation is primarily oral and independent of writing, literary pieces can also be used to improve pronunciation skills by having students read excerpts from them aloud in a meaningful way. This type of exercise also helps students understand how prosody is represented by punctuation marks and written accents (Calvo 2007). In short, practicing prosody by putting learners in contact with music and literature is a method of offering them the opportunity to understand cultural identities and build “complex world models” (Ferrús 2014), while simultaneously honing their L2 communicative skills.

8.5.11 Take advantage of technology

Technology can be very helpful in both classroom and bimodal environments (de-la-Mota 2004), allowing for didactic activities based on video games, animated agents, and treasure hunts, among other possibilities (Chun 2002; Padilla et al. 2009). While being fun, these tasks promote responsible and autonomous learning that can be combined with face-to-face teaching. Movie dubbing activities can also be stimulating, and a good way of practicing prosody in levels where reading is not considered an obstacle. First, learners become familiar with particular passages by paying attention to speech rate, as well as other relevant prosodic and contextual characteristics. Afterwards, they read the subtitles several times and then superimpose their voices over those of the characters in question (which play at a low volume). Finally, while learners read, the characters’ voices are silenced, after which these final recordings are compared to the models from the movie. This task easily allows for work in pairs or groups (Santamaría Busto 2007).

8.5.12 Combine verbal and non-verbal communication

Although human speech occurs mostly in the auditory-vocal modality, it is naturally blended with kinesics (see **kinesic behavior**), and multimodal communication involves this integrated speech system. In fact, adjusted gestural and visual expressions are at the foundation of the proprioceptive stimulation proposed by the Verbotonal methodology. McCafferty (2006) suggests that movement is a part of L2 acquisition that is related to prosody and that promotes automaticity and fluency. Certainly, multisensory reinforcements have been shown to be effective in enhancing acquisition. Zhang (2006), for instance, demonstrates the effectiveness of what she calls a *somatically-enhanced approach* to the acquisition of tones by L2 learners of Mandarin, and Kushch, Igualada, and Prieto (2015) emphasize the role of gesture, especially beats, in the teaching of L2 prosody to strengthen the memorization of vocabulary and linguistic structures, and to promote automaticity and increased

fluency. As such, in order to improve perception, it may be helpful to watch videos with short oral samples produced with natural gesture types while paying attention to the relationship between sound and gesture (and using slow motion if needed).

The production of prosody can be improved with imitation techniques. As Celce-Murcia, Brinton, and Goodwin (1996) explain, *shadowing* (or *echoing*) means trying to copy the pronunciation of a speaker with minimal delay (as in simultaneous interpretation). This technique has been shown to facilitate the learning of intonation and increase fluency (Hsieh, Dong, and Wang 2013, for a discussion, see Hamada 2017); however, the difficulty of perceiving immediate feedback and the probable lack of access to higher grammatical units endowed with meaning (which does occur when interpreting) are factors that require further investigation. A playful method of shadowing is karaoke. Another imitation technique, *mirroring*, involves emulating not only the pronunciation but also the kinesic behavior of a native speaker. Mirroring can be used to reinforce gestures that are consistent with spontaneous oral productions, especially those that have a role in accenting, phrasing, and cuing tonal events. One more possibility is working with audio-visually congruent speech; for example, after showing a short silent video with natural movements, learners could be asked to produce a congruent utterance using the gestures observed. Conversely, after listening only to the sound, learners could be asked to say the sequence again while including congruent gestures.

8.5.13 Work with the acoustic signal

There is a learning curve for teachers who desire to implement activities based on the acoustic signal; however, taking the time to learn how to navigate relevant tools is highly encouraged and recommended, as it opens the door to a side of speech that most students will have never seen. Recorders, signal displayers, sound analyzers and manipulators, and speech synthesizers offer the opportunity to productively utilize an audio signal, including those of students in real time, and to design activities tailored to work on specific prosodic features. These activities stimulate self-tutelage in a conscious way, and allow students to attempt to palliate the effects of “phonological deafness” through visual activity. They can be used in the classroom with some guidance or implemented as perception tasks, group activities, and creative games. Some programs (e.g., Audacity, Praat, and WaveSurfer) are reliable, free, based on cross-platform software, and not very hard to use (at least for the intended purposes). One should keep in mind, however, that not all acoustic data displayed by a pitch analyzer are relevant as visual feedback (de-la-Mota 2004): first, not all the information in a pitch analyzer is decodable or relevant for perception, second, the modification of some acoustic parameters is just a physiological consequence of having modified others, and third, the sound wave provides acoustic information about elements other than intonation, including segmental features and non-linguistic information; for instance, minor melodic variation (i.e., micro-melodic) in a fundamental frequency (F0) contour, which is just an effect of

particular sounds (e.g., **voiceless**), has no relevant impact on the intonational meaning of a whole utterance. Therefore, the display has to be interpreted in a careful and cautious manner. Regardless of these issues (and even though they require initial effort on the part of teachers), looking at prosodic factors in a software program is not overly complicated and is a fruitful exercise, as seen in Labrador and Fernández (1994), Llisterri (2003), de-la-Mota (2004), Wilson (2005), Campillos Llanos (2010), Martin (2010), Lahoz-Bengoechea (2011), Cabedo (2012), Gil Fernández (2012), Chun (2013) and Derwing and Munro (2015), among others. Some of the possibilities that these kinds of software provide for the teaching and learning of prosody are summarized below. They can be combined with each other and can also be integrated into other activities.

8.5.13.1 *Filtering and humming*

Filtering involves keeping a select band of frequencies of an utterance and removing all others. Hence, this procedure is, to some degree, similar to some functions of the **SUVAG equipment** used in the Verbotonal method. Furthermore, although a hum can be produced directly by the vocal tract, this technique not only allows us to work with the voices of learners, but also to listen to relevant parts of an utterance chosen from any sample of natural native speech. A low-pass filter is useful to obtain delexicalized speech, meaning it excludes most segmental information while maintaining prosodic characteristics of interest. Humming and voice modulation (including drama voice modulation techniques) can also be a way to work on production. As is clear from the results obtained by Ortega-Llebaria and Colantoni (2014), humming can be an effective starting point for practicing prosody since learners are much more accurate, both in production and perception, when tasks do not require them to access meaning through context. The use of natural and synthesized hums is very suitable for beginner learners (see the sample activity in the Appendix of this chapter).

8.5.13.2 *Manipulating the qualities of natural sound*

Each acoustic parameter can be studied and modified independently, thus allowing for multiple applications, such as exploring the perceptual effects of changing one or more parameters (e.g., pitch or duration and pitch together). First, the values of an acoustic parameter are altered, and then, the product is resynthesized to obtain a new utterance that sounds natural. Illustrations of this technique can be found in experimental works like Chen (2009), Lahoz-Bengoechea (2011), and Zárate-Sánchez (2015). It is useful because it allows for a comparison between learners' prosodic productions and modified versions, the latter of which more closely approximate the target. It is also an opportunity to artificially exaggerate (within limits) the characteristics of the target associated with differences found in learners' productions, which demonstrates a link between nuanced pronunciation (in the Verbotonal sense) and prosodic cues.

8.5.13.3 *Playing (and optionally displaying) selected parts of utterances*

Specific portions of utterances can be cut out very precisely, which facilitates the illustration of phenomena such as resyllabification, stress clash, initial pitch in questions, intonational boundaries, and non-silent pauses. This approach promotes working on intonation part-by-part, for instance, by cutting out different sized chunks of an utterance, starting with the nuclear configuration and going backwards (i.e., regressive segmentation), as recommended by Verbotonalists. To do this, the final part of an utterance is cut out and the length of cut portions of the utterance increases progressively in reverse order (from the end to the beginning) until the complete utterance is played (see the sample activity in the Appendix of this chapter and Chapter 13 of this volume).

8.5.13.4 *Morphing (transplanting or cloning) prosody*

In this technique, the values of some selected prosodic features are transferred from a production of a native speaker (i.e., the donor) to an utterance produced by a non-native learner (i.e., the recipient) (Jilka 2000). After this, a new sound is created through resynthesis. This creation can be used in self-imitation activities since the utterance generated is native-like and far from foreign accented speech. Such an exercise is effective because listening to one's own voice when it sounds like a closer approximation to the target is both surprising and stimulating (Yoon 2007; Pettorino and Vitale 2012; Pettorino, De Meo, and Vitale 2014).

8.5.14 *Play and display sounds, show schemae, and use prosodic transcription*

Teachers can also consider using visualization by playing a sound associated with a screenshot showing an acoustic analysis, a schematic drawing of an intonational contour or even an abstract (phonological) transcription; for instance, based on standard ToBI conventions for transcribing and annotating the prosody of speech, a new system for teaching and learning intonation was developed by Estebas-Vilaplana (2013). The TL_ToBI model (ToBI for Teaching and Learning) is designed to help students learn the most common tunes of English intonation in a distance-learning environment using audio and visual aids. It combines the metrical and tonal structure of the AM phonological framework with the tune descriptions and nuclear configurations of the British School. After some adaptation, the system could also be effective for other languages. In order to integrate it in an L2 course, teachers need to be aware that it involves conscious learning of the phonological system of the L2 in question (as well as knowledge of the L1 in question's system), and that they need experience with the transcriptional system.

8.5.15 *Become a facilitator*

To this point, with regard to incorporating prosody into L2 Spanish teaching practices, we have reviewed ways in which teachers can select appropriate material (e.g., different types of video, audio, etc.), devise activities tailored to the linguistic level and cultural interests of students, promote collaborative work, and stimulate oral and written communication. Additionally, teachers can encourage the use of other kinds of resources while providing learners with appropriate guidelines, such as Virtual Learning Environments (VLE) (or Learning Management Systems, LMS) and video/audio messaging with native speakers. Finally, the construction of a Personal Learning Environment (PLE) can help individual students develop a customized acquisition process (see Cassany 2014).

8.6 Final remarks

After having reviewed work on L2 Spanish prosody, as well as methods and materials dedicated to teaching it, we see that its current state seems both disappointing and inspiring; that is, further research is needed, available instructional materials are scarce, and there is a wide range of methodological proposals. An additional challenge that Padilla (2007) notes is that practicing oral skills in the classroom in greater depth would potentially involve curricular revisions related to hours of class time and course syllabi. Despite these issues, teachers need to do what they can to address the communicative importance of prosody with students of all levels. In order to make this task a little easier, this chapter has summarized the main characteristics of the prosodic system of Spanish from the point of view of its acquisition as an L2/FL, and has provided guidance for language teachers to consider how to prepare activities appropriate to their objectives, all while taking into account the theoretical underpinnings of Spanish's prosodic system.

To put the recommended suggestions into practice with students, it would be desirable to devote a brief portion of each class period to playful but meaningful pronunciation activities, preferably while connecting prosodic skills with other linguistic topics to be addressed in the rest of the class session. In the words of Cortés:

A nuestro modo de ver, lo propio es que el profesor tome el material en el punto en que el diseñador ha concluido su labor y a partir de ahí lo reprocese, reelabore, adapte, complete, depure, reenfoque, reorganice ... y lo ubique en el proceso instructivo, de modo que el grupo de aprendientes en cuestión pueda utilizarlo acorde con sus necesidades y expectativas.

(Cortés 2001, 139)

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Appendix

Sample activity

This sample didactic activity for prosody practice incorporates various elements discussed in Section 8.5. This particular version is aimed at students with minimal knowledge of Spanish; however, it can be adapted for use with higher levels of learners.

Activity

“Is this a question?”

Level

Absolute beginners (0–A1, novice low).

Objective

To introduce the perception of interrogative contours and learn the final low-rise contour of Spanish absolute yes–no questions used as information requests, a quite common (and unmarked) nuclear configuration.

Target learners

Those who use other final contours with the same pragmatic content in their L1, such as L1 Russian speakers (who mainly use a falling H*L L%). Those who use other intonational and lexical or syntactic strategies with the same communicative purpose can also benefit from the activity.

Time

Fifteen or 30 minutes, depending on the number of students and the activities selected.

Materials

Recordings and an audio/video player or a computer with Praat (Boersma and Weenink 2018).

Contextualization

Video, regressive segmentation, humming, and games are used to introduce the intonation of information-seeking yes–no questions.

Teacher preparation before class

1. Search for at least one video with a natural spontaneous conversation that contains an appropriate context for informative questions and make it available to

the class. Avoid fiction. Only the scene with the selected dialogue will be needed. Make sure the selected conversation is of good sound quality and that the native target language speaker producing the yes–no question(s) is clearly visible. Look for sentences ending in a paroxytonic word, where the stressed syllable will carry a low tone (L*) and the post-tonic syllable will have a rising boundary tone (HH%). Interrogatives of about eight or nine syllables would be fine to start with since this length is considered to be the average for an international unit between pauses (i.e., *el grupo fónico*, Navarro Tomás 1944). Some examples that meet these criteria are *¿La visitará en enero?* and *¿Llegará de madrugada?*

2. Manipulate the sound with a low-pass filter. Praat can be used for this procedure. First, read the sound file in Praat; it will be listed as a sound object in the left-hand column. Second, select the sound object and then the “view & edit” option. The sound analysis window will then be displayed. Third, open the “pitch” menu and select “extract visible contour.” A new pitch object will be created and listed in the left-hand column. Fourth, select it and open the “sound” menu. Fifth, select the option “to sound (hum).” The newly created sound object appears in the left-hand column and can be played and analyzed with the “view & edit” option.
3. Prepare all the short clips of interest before class by cutting out the selected portions through the “view & edit” option and saving them. The sound can also be played part by part in the classroom, but this requires knowing the exact start and end time of each fragment of interest.

Procedure

1. Introduce the activity. Explain the type of sentence of interest and its pragmatic nuance.
2. Practice perception.
 - 2.1. Explain the communicative situation of the video.
 - 2.2. Play the video. Have students pay attention to prosody and kinesic behavior.
 - 2.3. Play the audio only in the form of the entire filtered sentence. Start to work on regressive segmentation, that is, play small parts of the sentence, starting at the end and moving backwards. Play the last syllable first (e.g., *da* in *madrugada* from the example *¿Llegará de madrugada?*) to show the fast, high rising intonation, and then move on to the whole nuclear contour, starting from the beginning of the last accented syllable (e.g., *gada* in *madrugada*). You can use the syllable as a relevant unit to create **gates** (e.g., *da*, *gada*, *drugada*, *madrugada*, etc.) or you can work with higher prosodic units. The stimuli will continue to get longer until the sentence is played in its entirety.
 - 2.4. Optionally, introduce a fun activity, such as “musical statues” or “freeze dance.” All the students stand up, and when a stimulus starts, they have

to walk. As soon as the stimulus ends, they have to freeze. After playing the whole sentence a few times, begin interspersing short and long fragments so that the game becomes difficult. In the end, the stimuli will have been reproduced a lot of times and students will have been paying a lot of attention to them.

- 2.5. Finally, return to the natural context and play the video with all relevant linguistic information once again.
3. Practice production. Since each sentence will be short, even beginners will be able to emulate it, but if the preference is to avoid segmental information in the first days, students can just hum the prosodic content rather than producing the sentence normally. Regardless, it is important to require that they use appropriate nonverbal cues.

A game that can accompany this step is “ball toss,” in which a sentence is “passed” quickly from one student to another, who, in turn, will “pass” the same sentence to the next student after having answered *yes* or *no* with a nod. This gesture helps practice the informative nuance of the question.

4. Feedback. At the end, explain the linguistic purpose of and provide commentary about the session. Highlight the crucial role of intonation in *yes–no* questions due to the syntactic properties of Spanish.

Some considerations for teachers

1. Before
 - 1.1. First of all, which variety of Spanish is going to be taught to the learners? Geolectal variation on Spanish *yes–no* questions was presented in Section 8.2.
 - 1.2. If using software to obtain a hum is not an option, the teacher’s voice can be used to hum small units, full intonation units, or long passages.
 - 1.3. If obtaining samples from natural conversations is not possible, elicit data from a native speaker. Find at least two examples, such as *¿La visitará en enero?* and *¿Llegará de madrugada?*. It is best if a communicative interaction can be video-recorded.
2. During
 - 2.1. Avoid writing out text to teach prosody. Leave this for advanced learners of pronunciation.
 - 2.2. It is fine to reinforce the input by displaying its corresponding pitch contour, but not other acoustic information (unless students have previous acoustic training).
 - 2.3. Showing schematic contours can be helpful. One method of doing this is drawing the portion to which students are listening.
3. After
 - 3.1. Briefly review the learned contour during the next class session (5 minutes). Use a short video as example of the natural communicative interaction and play the filtered sentence a couple of times.

- 3.2. Continue working with interrogatives of different lengths, including those with just one syllable (e.g., *¿Van?*)
- 3.3. Pictures and realia can be used to work with interrogatives that include a certain stress pattern (i.e., forming questions about a certain object), as in *¿Le regalaron la lámina?*
- 3.4. As soon as this nuclear contour and some basic vocabulary and grammatical structures become familiar to students, the contour can be practiced in more complex games and in meaningful collaborative tasks (e.g., the “who is who” game and gymkhanas), which, in turn, can be used to work on sociocultural factors.
- 3.5. Work on the beginning of absolute yes–no questions used as information requests. As for the segmentation technique explained here, which is inspired by the Verbotonal method, it can be used in a similar way in the opposite direction. Progressive segmentation is preferred if the contour of interest is at the beginning of an utterance (i.e., initial prenuclear pitch accent), which is different in declaratives and interrogatives (Face 2007). As for Castilian Spanish yes–no questions, it is important to show that the initial pitch accent is high (although it could be unmarked) and that the most common contour seems to be the bitonal low–high. Practicing this could be done by playing a game in which a student produces part of the beginning of a sentence and another one has to finish it in one of an infinite number of ways.
- 3.6. Continue working with interrogative sentences, including vocatives or parenthetical comments (*¿Le regalaste, amigo mío, lo que habíamos comentado?*), especially with advanced learners (C1–C2, advanced high).

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